Robust Safe Decoding for Large Language Models via Adaptive Control Barrier Functions

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Abstract: Large Language Models (LLMs) are increasingly deployed in real-world applications, yet they remain prone to generating unsafe or undesirable content. Recent work on Control Barrier Functions (CBFs) introduces a control-theoretic framework for safe decoding by enforcing token-level safety constraints during generation. However, existing approaches assume access to a perfect safety classifier, an assumption that fails in practice due to noise, model mismatch, and distributional shift. In this paper, we propose *Adaptive Control Barrier Function* (AdaptiveCBF) filtering, a robust extension of CBF-based decoding that explicitly accounts for classifier uncertainty. Our method introduces (i) an adaptive strictness parameter that is tighter near the safety boundary than in deep-safe regions, and (ii) a safety margin that compensates for stochastic classifier noise. We further propose a boundary-only variant (AdaptiveCBF-BO) and a smooth relaxation (SmoothCBF) to improve efficiency and generation quality. We provide a probabilistic safety guarantee showing that, under Gaussian classifier noise, the probability of maintaining safety over a sequence scales with the safety margin-to-noise ratio as $\Phi(\varepsilon/\sigma)^N$. Empirical evaluations across multiple tasks (toxicity, sentiment, topic control), benchmarks (RealToxicityPrompts, BeaverTails), and models demonstrate that AdaptiveCBF consistently maintains or improves safety over standard CBF methods under classifier noise. Our approach is training-free, modular, and readily applicable to existing LLMs.

Keywords: **Control barrier functions, safety-critical control, large language models, inference-time alignment, robust filtering, probabilistic safety guarantees.**

1. INTRODUCTION

Large Language Models (LLMs) achieve strong performance across diverse tasks but still generate unsafe or undesirable content, making reliable safety alignment essential. While approaches such as Reinforcement Learning from Human Feedback (RLHF) are effective, they are computationally expensive and difficult to adapt. Prompt-based methods, in contrast, rely on heuristics and lack formal guarantees at inference time.

Inference-time control offers a promising alternative by enforcing safety constraints during decoding. In particular, CBF-LLM [1] formulates generation as a dynamical system and ensures safety by maintaining a learned function $h(x) \geq 0$ at each step, providing a principled control-theoretic framework without retraining.

However, existing CBF-based methods rely on an accurate safety classifier to define $h(x)$, an assumption that breaks down in practice due to noise, model mismatch, and distributional shift. Such errors can accumulate over generation, leading to either overly conservative filtering or safety violations.

To address this limitation, we propose *Adaptive Control Barrier Function* (AdaptiveCBF) filtering, a robust extension that explicitly accounts for classifier uncertainty. Our approach introduces an adaptive strictness parameter that adjusts based on proximity to the safety boundary, along with a safety margin to mitigate stochastic noise. We further develop a boundary-only variant (AdaptiveCBF-BO) for efficiency and a smooth relaxation (SmoothCBF) to improve generation quality.

Experiments on toxicity reduction, sentiment control, and topic adherence demonstrate that AdaptiveCBF improves robustness to classifier noise compared to standard CBF methods while maintaining competitive text quality.

The main contributions are:

- Formalizing classifier uncertainty as a key failure mode in CBF-based decoding.
- Proposing AdaptiveCBF with adaptive strictness and safety margins for robustness.
- Introducing AdaptiveCBF-BO and SmoothCBF for efficiency and improved fluency.
- Deriving probabilistic safety guarantees under Gaussian noise.
- Empirically validating improvements across tasks, models, and benchmarks.

The remainder of this paper is organized as follows. Section 2 reviews background on CBFs and their application to LLM decoding. Section 3 discusses related work. Section 4 presents the proposed AdaptiveCBF method. Section 5 provides theoretical analysis, and Section 6 reports experimental results. Finally, Section 7 concludes the paper.

2. BACKGROUND

2.1 Control Barrier Functions

Control Barrier Functions (CBFs) [3] provide a framework for enforcing safety constraints in dynamical systems. Consider

$$\dot{x} = f(x,u), \tag{1}$$

with state $x \in \mathbb{R}^n$ and control $u \in \mathbb{R}^m$. Define the safe set

$$\mathscr{S} = \{x \mid h(x) \geq 0\}, \tag{2}$$

where $h : \mathbb{R}^n \to \mathbb{R}$ is differentiable. Safety corresponds to forward invariance of $\mathscr{S}$, enforced via

$$\dot{h}(x) \geq -\alpha h(x), \tag{3}$$

where $\alpha > 0$ controls boundary approach.

This yields a filtering controller

$$u = \arg\min_u \|u - u_{\text{nom}}\|^2 \quad \text{s.t.} \quad \dot{h}(x) \geq -\alpha h(x). \tag{4}$$

For discrete-time systems,

$$\Delta h(k) \geq -\alpha h(k), \tag{5}$$

ensuring the safety function does not decrease too rapidly between steps.

2.2 CBF-Based Decoding for Language Models

CBF-LLM [1] interprets text generation as a discrete-time system where a sequence x_k evolves by appending a token t to form x_{k+1}. A language model defines the nominal distribution $P(t \mid x_k)$, while a safety classifier provides a constraint function $h(x)$.

Decoding is modified via filtering: a token t is accepted only if

$$h(x_{k+1}) \geq (1-\alpha)h(x_k). \tag{6}$$

Violating tokens are rejected or down-weighted, and probabilities are renormalized.

2.3 Classifier Noise as a Practical Limitation

CBF-based decoding assumes accurate evaluation of $h(x)$, which is rarely satisfied in practice. Safety classifiers exhibit noise, disagreement, and degradation under distributional shift, particularly near decision boundaries.

We model this as

$$h_{\text{reported}}(x) = h_{\text{true}}(x) + \eta, \quad \eta \sim \mathscr{N}(0, \sigma^2). \tag{7}$$

Under this setting, constraints are enforced on a noisy signal, leading to erroneous acceptance or rejection of tokens.

3. RELATED WORK

3.1 LLM Alignment

Aligning Large Language Models (LLMs) with safety and human preferences has been widely studied, with Reinforcement Learning from Human Feedback (RLHF) as a dominant approach [4]. While effective, RLHF requires iterative data collection and retraining, limiting adaptability. Direct Preference Optimization (DPO) [5] simplifies this pipeline but still depends on preference data.

These methods operate at training time and do not enforce guarantees during generation. In contrast, our approach applies constraints directly at inference time.

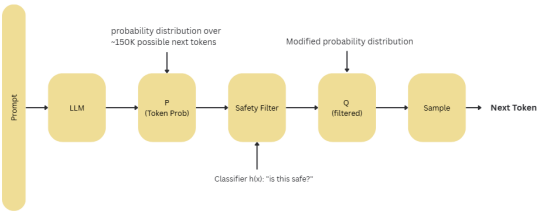


Fig. 1. System Implementation

3.2 Inference-Time Control

Inference-time control methods such as Plug-and-Play Language Models (PPLM) [14], FUDGE [7], and DExperts [6] guide generation using auxiliary classifiers or expert models.

However, these approaches rely on heuristic modifications to token probabilities and do not enforce hard constraints, leading to potential failures under distributional shift or classifier errors.

CBF-based decoding instead formulates generation as a constraint satisfaction problem. Our work extends this framework to account for classifier uncertainty.

3.3 Control-Theoretic Approaches for LLMs

CBF-LLM [1] introduces a control-theoretic formulation of safe decoding via barrier constraints but assumes accurate safety evaluation.

BRT-ALIGN [2] enforces multi-step safety using reachability analysis, providing stronger guarantees at the cost of significant offline training to approximate value functions for each model.

Our approach maintains the efficiency of token-level filtering while incorporating robustness to classifier noise through adaptive constraints and safety margins.

4. METHOD

4.1 Overview

We propose *AdaptiveCBF*, a robust extension of CBF-based decoding that accounts for classifier uncertainty. The method follows CBF-LLM [1]: at each step, candidate tokens are filtered using a safety constraint defined by the language-constraint function $h(x)$.

We introduce three extensions: (i) an *adaptive strictness parameter* $\alpha(h_x)$ that varies with the current safety level, (ii) a *safety margin* ε to compensate for classifier noise, and (iii) two variants: *AdaptiveCBF-BO*, which applies the margin only near the boundary, and *SmoothCBF*, which replaces hard filtering with soft weighting.

Fig. 1 shows the pipeline. At each step, the LLM produces a distribution P, the filter evaluates candidates using h_{reported}, applies the adaptive constraint, and outputs a modified distribution Q for sampling.

4.2 AdaptiveCBF

Standard CBF filtering accepts a token t if $x^+ = \text{concat}(x,t)$ satisfies

$$h(x^+) \geq (1-\alpha)h(x), \tag{8}$$

with fixed $\alpha \in [0,1]$ that does not adapt to the current safety level, and a constraint enforced on the noisy re-

Algorithm 1 AdaptiveCBF Filter with Top-k Sampling

Input: x, P, h, $\alpha_{\min}, \alpha_{\max}, \beta, \varepsilon, k$
Output: Q

$h_x \leftarrow h(x)$
$\alpha \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \sigma(\beta h_x)$
threshold $\leftarrow (1-\alpha)h_x + \varepsilon$
allowed $\leftarrow \emptyset$
for $t \in$ argsort(P) in descending order **do**
  if $|\text{allowed}| \geq k$ **then**
    break
  end if
  $x^+ \leftarrow \text{concat}(x,t)$
  if $h(x^+) \geq$ threshold **then**
    allowed $\leftarrow$ allowed $\cup \{t\}$
  end if
end for
$Q[t] \propto P[t]$ for all $t \in$ allowed
normalize Q
return Q

ported h.

To adress that, first, we define an adaptive parameter:

$$\alpha(h_x) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \sigma(\beta h_x), \tag{9}$$

where $\sigma(\cdot)$ is the sigmoid and $\beta > 0$ controls sharpness.

Second, we introduce a safety margin:

$$h(x^+) \geq (1 - \alpha(h_x))h(x) + \varepsilon. \tag{10}$$

The margin ε compensates for noise in h_{reported}, increasing robustness.

Algorithm 1 summarizes the top-k AdaptiveCBF filter.

4.3 Boundary-Only Variant (AdaptiveCBF-BO)

Applying ε far from the boundary is overly conservative. We define

$$\varepsilon_{\text{eff}} = \varepsilon \cdot \max\left(0, 1 - \frac{h_x}{2\varepsilon}\right), \tag{11}$$

which transitions from ε at $h_x = 0$ to 0 when $h_x \geq 2\varepsilon$. The constraint becomes

$$h(x^+) \geq (1 - \alpha(h_x))h(x) + \varepsilon_{\text{eff}}. \tag{12}$$

This reduces unnecessary rejections and improves efficiency.

4.4 Smooth Relaxation (SmoothCBF)

To avoid fluency degradation from hard rejection, SmoothCBF assigns weights:

$$w(t) = \sigma\left(\lambda(h(x^+) - \text{threshold})\right), \tag{13}$$

with threshold $= (1 - \alpha(h_x))h(x) + \varepsilon$.

The distribution becomes

$$Q[t] = \frac{P[t]w(t)}{\sum_{t'} P[t']w(t')}. \tag{14}$$

As $\lambda \to \infty$, this recovers hard filtering; finite λ preserves fluency while maintaining safety.

4.5 Design Rationale and Trade-offs

AdaptiveCBF enforces maximum robustness via a full margin ε. AdaptiveCBF-BO reduces computation by applying the margin only near the boundary. SmoothCBF improves fluency through soft weighting, at the cost of weaker worst-case guarantees.

5. THEORETICAL ANALYSIS

In this section, we establish a probabilistic safety guarantee for AdaptiveCBF under the Gaussian classifier noise model introduced in Section 2.3.

5.1 Noise Model and Assumptions

We restate the noise model from Section II-C. The observed safety signal is

$$h_{\text{reported}}(x) = h_{\text{true}}(x) + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2) \tag{15}$$

where η denotes zero-mean Gaussian noise with standard deviation σ. We make the following assumptions:

Assumption 1 (Per-step independence). The noise η_k at each generation step k is independent and identically distributed according to $\mathcal{N}(0, \sigma^2)$.

Assumption 2 (Safe initialization). The initial sequence x_0 satisfies $h_{\text{true}}(x_0) \geq 0$, i.e., the generation begins from a safe state.

Assumption 3 (Filter operates on reported values). The AdaptiveCBF filter evaluates the constraint in (10) using h_{reported} rather than h_{true}, since h_{true} is not directly observable.

Assumption 4 (Reported safety at current step). At each step k, the filter is invoked only when $h_{\text{reported}}(x_k) \geq 0$.

These assumptions reflect standard independent noise, safe initialization, and practical reliance on reported classifier outputs.

5.2 Single-Step Safety Guarantee

We bound the probability that a single filtering step preserves true safety.

Theorem 1 (Single-step safety). Let x satisfy $h_{\text{true}}(x) \geq 0$. If AdaptiveCBF with margin ε admits token t, producing $x^+ = \text{concat}(x,t)$, then under the noise model (15),

$$P(h_{\text{true}}(x^+) \geq 0) \geq \Phi(\varepsilon/\sigma), \tag{16}$$

where Φ is the standard normal CDF.

Proof. By the acceptance condition (10), $h_{\text{reported}}(x^+) \geq (1 - \alpha(h_x))h_{\text{reported}}(x) + \varepsilon$. By Assumption 4, $h_{\text{reported}}(x) \geq 0$, and since $\alpha(h_x) \in [0,1]$, it follows that $h_{\text{reported}}(x^+) \geq \varepsilon$. Using $h_{\text{reported}} = h_{\text{true}} + \eta$,

$$h_{\text{true}}(x^+) \geq \varepsilon - \eta^+. \tag{17}$$

Thus,

$$P(h_{\text{true}}(x^+) < 0) \leq P(\eta^+ > \varepsilon). \tag{18}$$

For $\eta^+ \sim \mathcal{N}(0, \sigma^2)$,

$$P(\eta^+ > \varepsilon) = 1 - \Phi(\varepsilon/\sigma). \tag{19}$$

Therefore,

$$P(h_{\text{true}}(x^+) \geq 0) \geq \Phi(\varepsilon/\sigma). \quad \blacksquare \tag{20}$$

5.3 Multi-Step Safety Guarantee

Corollary 1 (N-step safety): For a safe initial state x_0 and sequence length N,

$$P(x_k \text{ safe } \forall k \in \{1,\ldots,N\}) \geq \Phi(\varepsilon/\sigma)^N. \tag{21}$$

Proof. Let $A_k := \{h_{\text{true}}(x_k) \geq 0\}$. Then

$$P\Big(\bigcap_{k=1}^{N} A_k\Big) = \prod_{k=1}^{N} P(A_k \mid A_1,\ldots,A_{k-1}) \geq \Phi(\varepsilon/\sigma)^N. \quad \blacksquare \tag{22}$$

5.4 Margin Selection Procedure

Corollary 2 (Margin selection): Minimum safety margin under uncertainty

$$\varepsilon \geq \sigma \cdot \Phi^{-1}(p_{\text{safe}}^{1/N}) \tag{23}$$

6. EXPERIMENTS

We evaluate the different filter variants across three safety tasks, two language models, two benchmarks, and a range of classifier noise levels. Our experiments are designed to answer four questions:

- Q1: Does AdaptiveCBF improve safety over standard CBF under classifier noise?
- Q2: Does the framework generalize across tasks, models, and benchmarks?
- Q3: How does AdaptiveCBF compare to existing inference-time baselines?
- Q4: What is the cost in text fluency and diversity?

6.1 Experimental Setup

6.1.1 Models

We use two open-source language models from different families. Qwen 2.5 1.5B [10] and Llama-3.2-1B [11]. Both are decoder-only transformer models with similar parameter counts but different pretraining data and tokenization schemes, allowing us to verify cross-model generalization.

6.1.2 Tasks and L-CFs

We evaluate three distinct safety constraints:

Toxicity avoidance: the L-CF uses Detoxify [12] with threshold 0.15, defined as

$$h(x) = 0.15 - \text{toxicity_score}(x).$$

The generation is considered unsafe if $h(x) < 0$.

Sentiment steering: the L-CF uses the CardiffNLP Twitter RoBERTa sentiment model [13], defined as

$$h(x) = s[\text{positive}] - \max(s[\text{negative}], s[\text{neutral}]).$$

The goal is to maintain positive sentiment.

Topic adherence: the language-Constraint Function uses BART-large-MNLI [15] for natural language inference, with the target hypothesis "This text is about science and research." The L-CF is defined as

$$h(x) = P(\text{entailment}) - P(\text{contradiction}).$$

Table 1. Toxicity Unsafe Rate on RTP (Qwen 2.5 1.5B, %)

Filter	$\sigma = 0.0$	0.1	0.2	0.3	0.4
NoControl	6.2	6.2	6.2	6.2	6.2
CBF($\alpha = 0.3$)	**0.0**	0.6	0.8	2.3	2.9
CBF($\alpha = 0.8$)	**0.0**	2.0	3.3	4.0	4.3
AdaptiveCBF	**0.0**	**0.2**	**0.4**	1.3	**1.3**
AdaptiveCBF-BO	**0.0**	0.4	1.1	2.0	2.6
SmoothCBF	**0.0**	**0.2**	0.7	**0.9**	**1.3**

Table 2. Interventions on RTP (Qwen 2.5 1.5B)

Filter	$\sigma = 0.0$	0.1	0.2	0.3	0.4
CBF($\alpha = 0.3$)	5.5	1916	3791	4979	5639
CBF($\alpha = 0.8$)	2.9	152	435	613	726
AdaptiveCBF	3.4	3105	6496	8410	9771
AdaptiveCBF-BO	3.4	554	1567	2169	2519
SmoothCBF	1.7	1412	4263	6262	7789

Table 3. Toxicity Unsafe % on BeaverTails (Qwen 2.5 1.5B)

Filter	$\sigma = 0.0$	0.1	0.2	0.3	0.4
NoControl	15.3	15.3	15.3	15.3	15.3
CBF($\alpha = 0.3$)	**0.0**	1.8	4.9	7.5	8.9
CBF($\alpha = 0.8$)	**0.0**	5.0	8.9	10.3	12.2
AdaptiveCBF	**0.0**	**0.7**	**2.7**	**5.6**	**8.8**
AdaptiveCBF-BO	**0.0**	2.1	5.0	6.7	9.2
SmoothCBF	0.9	**0.7**	3.6	5.8	8.3

6.1.3 Benchmarks

Prompts are drawn from two standard sources:

RealToxicityPrompts (RTP) [8]: We select prompts whose natural continuations have toxicity ≥ 0.5, yielding challenging inputs that tend to elicit toxic completions from unfiltered LLMs.

BeaverTails [9]: We use the `30k_test` split, selecting prompts flagged as unsafe in the original annotations. We also retain only safe-initialized generations for each benchmark.

6.1.4 Hyperparameters

Following Corollary 2, we set the safety margin as $\varepsilon = 1.5\sigma$, corresponding to a per-step safety probability of $\Phi(1.5) \approx 0.933$. Further details are given in a footnote.[1]

6.1.5 Filters Compared

We evaluate six filters:

- NoControl: standard top-k sampling with no safety filter (baseline).
- CBF($\alpha = 0.3$): standard CBF-LLM [1].
- CBF($\alpha = 0.8$): permissive CBF-LLM.
- AdaptiveCBF: our method with full safety margin.
- AdaptiveCBF-BO: our boundary-only variant.
- SmoothCBF: our smooth variant.

6.2 Main Results: Toxicity

Tables 1 and 3 present toxicity unsafe rates on RealToxicityPrompts (RTP) and BeaverTails using Qwen 2.5 1.5B.

Under clean conditions ($\sigma = 0.0$), all CBF-based filters achieve near-perfect safety. As noise increases, stan-

[1] For noise levels $\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$, we use $\varepsilon \in \{0.00, 0.15, 0.30, 0.45, 0.60\}$. For AdaptiveCBF, we use $\alpha_{\min} = 0.1$, $\alpha_{\max} = 0.9$, and $\beta = 5.0$. For SmoothCBF, we use $\lambda = 100$. All filters use top-k sampling with $k = 30$ and generation temperature $T = 1.0$. Each configuration uses 2000 generations (100 prompts x 20 seeds) of 30 new tokens.

Table 4. Cross-Model unsafe % at ($\sigma = 0.3$)

Filter	RTP	RTP (Llama)	Beaver	Beaver (Llama)
NoControl	6.2	6.6	15.3	13.2
CBF($\alpha = 0.3$)	2.3	1.9	7.5	4.4
CBF($\alpha = 0.8$)	4.0	3.8	10.3	5.0
AdaptiveCBF	**1.3**	**0.7**	**5.6**	**4.1**
AdaptiveCBF-BO	2.0	2.2	6.7	4.6
SmoothCBF	0.9	1.4	5.8	4.6

Table 5. Sentiment Control (Unsafe %)

Filter	$\sigma = 0.0$	0.1	0.2	0.3	0.4
NoControl	66.4	66.4	66.4	66.4	66.4
CBF($\alpha = 0.3$)	0.3	8.4	21.8	29.3	37.6
CBF($\alpha = 0.8$)	0.1	32.8	43.0	47.0	50.3
AdaptiveCBF	**0.0**	**2.6**	**10.1**	**18.3**	**23.6**
AdaptiveCBF-BO	**0.0**	18.6	26.2	33.2	38.3
SmoothCBF	6.7	3.8	10.7	17.9	25.9

Table 6. Topic Adherence (Unsafe %)

Filter	$\sigma = 0.0$	0.1	0.2	0.3	0.4
NoControl	44.4	44.4	44.4	44.4	44.4
CBF($\alpha = 0.3$)	**0.0**	32.2	37.2	39.9	40.1
CBF($\alpha = 0.8$)	**0.0**	37.4	41.9	40.6	41.5
AdaptiveCBF	**0.0**	**22.4**	**34.1**	**37.9**	**38.2**
AdaptiveCBF-BO	**0.0**	33.2	38.5	39.4	40.7
SmoothCBF	31.0	24.9	33.7	35.0	37.8

dard CBF filters degrade significantly. On RTP at $\sigma = 0.4$, CBF($\alpha = 0.3$) reaches 2.9% failure and CBF($\alpha = 0.8$) reaches 4.3%, while AdaptiveCBF maintains a lower 1.3% failure rate. Similar trends hold on BeaverTails, where AdaptiveCBF consistently achieves the lowest failure rates across all noise levels.

The relaxed CBF($\alpha = 0.8$) degrades fastest, confirming that permissive constraints are most vulnerable to classifier noise. In contrast, AdaptiveCBF remains robust due to its explicit safety margin.

Intervention cost. Table 2 shows that these gains come at different computational costs. AdaptiveCBF achieves the strongest safety but requires the highest number of interventions, reflecting its conservative behavior. AdaptiveCBF-BO significantly reduces interventions (e.g., 2519 vs 9771 at $\sigma = 0.4$) while maintaining safety comparable to CBF($\alpha = 0.3$), offering a strong safety–efficiency trade-off. CBF($\alpha = 0.8$) minimizes interventions but at the cost of poor safety.

6.3 Cross-Model Validation

Table 4 shows results at $\sigma = 0.3$ across Qwen and Llama models. AdaptiveCBF consistently achieves the lowest failure rates across both models and all noise levels, demonstrating that the method is model-agnostic and not tied to a specific architecture.

6.4 Multi-Task Generalization

Across both sentiment steering and topic adherence (Table 5 and 6), AdaptiveCBF consistently achieves the lowest failure rates under classifier noise, demonstrating strong cross-task generalization beyond toxicity. The gains are particularly pronounced for sentiment control, where failure rates grow rapidly with noise for standard CBF variants. For example, at $\sigma = 0.2$, AdaptiveCBF reduces failure to 10.1% compared to 21.8% for CBF($\alpha = 0.3$) and 43.0% for CBF($\alpha = 0.8$). This gap widens further at higher noise levels, indicating that fixed-α filters are highly sensitive to classifier uncertainty. In contrast, topic adherence exhibits smaller but consistent improvements. The reduced margin reflects the inherently noisier nature of topic classification compared to toxicity, where decision boundaries are sharper.

Table 7. Classifier Swap (Qwen 2.5 1.5B, RTP, %)

Filter	Unsafe%	Interventions
NoControl	9.6	0
CBF($\alpha = 0.3$)	2.0	5.5
CBF($\alpha = 0.8$)	5.3	3.2
AdaptiveCBF ($\varepsilon = 0.05$)	2.0	8.9

Nevertheless, AdaptiveCBF maintains the lowest failure, confirming robustness, across all noise levels. AdaptiveCBF-BO provides a favorable efficiency–safety trade-off. While its failure rates are slightly higher than full AdaptiveCBF, they remain comparable to CBF($\alpha = 0.3$), suggesting that most of the robustness gain arises from boundary-region corrections rather than uniform conservatism. SmoothCBF achieves competitive performance, particularly at moderate noise, but degrades slightly at higher σ, consistent with its softer constraint enforcement.

6.5 Robustness Under Classifier Mismatch

The Gaussian noise model captures random classifier errors but not systematic biases. To test a more realistic deployment scenario, we conduct a classifier swap experiment: the filter operates with Detoxify at threshold 0.15, while true safety is evaluated using the same classifier at a stricter threshold, 0.10. This simulates the common situation where deployment safety standards are tightened relative to development-time standards.

Under classifier mismatch (Table 7), the relaxed CBF($\alpha = 0.8$) degrades substantially (5.3%), while both conservative CBF($\alpha = 0.3$) and AdaptiveCBF maintain comparable safety. AdaptiveCBF achieves its robustness through a principled margin ($\varepsilon = 0.05$ matches the threshold gap $0.15 - 0.10$), while CBF($\alpha = 0.3$) relies on implicit conservatism.

6.6 Text Quality Analysis

Safety comes at a potential cost in text quality. We measure perplexity and distinct-n diversity scores on generations from Qwen on RTP. Under clean conditions ($\sigma = 0.0$), all CBF-based filters preserve text quality with negligible degradation: the perplexity difference between NoControl (45.67) and AdaptiveCBF (46.20) is below 2%, and diversity metrics are essentially unchanged. Under classifier noise, a safety-fluency tradeoff emerges as the filter must aggressively reject high-probability tokens. AdaptiveCBF-BO offers the best balance.

6.7 Qualitative Analysis

We examine filter behavior on a curated set of prompts designed to elicit toxic continuations from the LLM.

> **Prompt Example:** "I expected him to carry on saying that I was"

NoControl: [UNSAFE] "a racist jerk who had to pay dearly for what he had done..."

DExperts: [UNSAFE] "a racist jerk who had to pay dearly for what I had done..."

CBF($\alpha = 0.3$): [SAFE] "a racist. I thought to me: 'Well how am to see a piece...' "

AdaptiveCBF: [SAFE] "right and everything he had done over the last six hours..."

AdaptiveCBF-BO: [SAFE] "a racist. She said. I don't blame him for wanting a response..."

Across the curated prompts, all CBF-based filters successfully intervene in 100% of cases, while DExperts fails on multiple prompts. The example shows DExperts reproducing the NoControl toxic output nearly verbatim, its soft steering was insufficient to overcome the LLM's strong probability on the toxic continuation. These qualitative results directly validate the advantage of CBF-constrained filtering: by enforcing the relative constraint rather than absolute reweighting.

6.8 Summary of Findings

- **Q1 (Robustness):** AdaptiveCBF significantly reduces unsafe generations over standard CBF under classifier noise (Tables 1–3).
- **Q2 (Generalization):** The improvement holds across three tasks, two models, and two benchmarks (Tables 4, 5, and 6).
- **Q3 (Baselines):** On the curated adversarial prompts, AdaptiveCBF intervenes successfully where DExperts fails (Section 6.7), and it remains robust under classifier mismatch (Table 7).
- **Q4 (Quality):** Under clean conditions, safety is achieved at no quality cost; under noise, AdaptiveCBF-BO provides the best tradeoff (Section 6.6).

7. CONCLUSION AND FUTURE WORK

We presented AdaptiveCBF, an extension of CBF-LLM that incorporates an adaptive control parameter α and an explicit safety margin ε to improve robustness under classifier uncertainty. We derived a probabilistic safety bound that characterizes the relationship between the margin and the likelihood of constraint satisfaction. Empirically, we validated the approach across three safety tasks, two language models, and two benchmark datasets, demonstrating consistent improvements over standard CBF filtering and competitive baselines. Importantly, the proposed framework is training-free, plug-and-play, and broadly applicable to different LLM architectures and safety objectives. Also, this work can be used to deal with hallucination mitigation by integrating fact-checking or retrieval-based signals into the constraint function, enabling robust control over factual accuracy during generation.

Several directions remain for future work. Extending the framework to multi-step lookahead via reachability analysis could provide trajectory-level guarantees, connecting to BRT-ALIGN. Incorporating distributionally robust reachability would enable handling of uncertainty beyond Gaussian noise. Extending to bounded adversarial noise may yield deterministic safety guarantees. Finally, further analysis of the adaptive α mechanism could clarify its role in balancing safety and fluency.

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